Spatial-Behavioural Analytics: Mapping Nutritional Vulnerability and Food Accessibility in Singapore's Aging Districts

SC2320 Data Analytics and Mining, Final Project Report

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Abstract

Singapore is ageing rapidly and by 2030 roughly one in four residents will be aged sixty five or above. Elderly residents face compounding barriers to food security when sparse affordable food venues, limited public transport and inadequate barrier free infrastructure coincide in the same neighbourhood, yet these dimensions are almost never measured jointly. This project presents a reproducible, end to end data mining pipeline that uses ten publicly available Singapore datasets at the Urban Redevelopment Authority subzone level, and applies three connected analytical phases to 219 residential subzones. Phase 1 runs Apriori association rule mining on a transparent twelve item binary matrix. Phase 2 compares K-Means, Agglomerative Hierarchical Clustering with Ward linkage and DBSCAN using a mean rank protocol over silhouette, Davies-Bouldin and Calinski-Harabasz metrics, and validates the chosen partition with a bootstrap Adjusted Rand Index stability analysis. Phase 3 trains a Random Forest against a documented proxy vulnerability target, benchmarks it against Logistic Regression, a depth four Decision Tree and a majority class floor, and uses Gini feature importance to explain what drives the risk signal. The pipeline narrows 332 subzones to a ranked list of 20 priority hotspots with a three layer trail: co-occurrence rules, a neighbourhood typology and a continuous vulnerability score. All outputs can be regenerated from the git repository by a single command.

1 Introduction

1.1 Motivation

Singapore is one of the fastest ageing societies in Asia. According to the Department of Statistics, roughly one in four residents is projected to be aged sixty five or above by 2030. This is not simply a demographic headline. It is a spatial planning challenge, because elderly residents interact with the built environment in ways that are different from working age adults. Three of those interactions matter particularly for this project.

The first is affordable prepared food. Hawker centres, wet markets and supermarkets are the primary food supply for older residents who cook less frequently and have lower average incomes than working age cohorts. Walking access to these venues is not a convenience but a daily necessity for residents who do not drive.

The second is accessible public transport. The density of bus stops and MRT exits within a short walk of a subzone determines whether residents who cannot walk far can still reach food, healthcare and social services.

The third is barrier free built infrastructure. Even where food venues and transit exist, steep approaches, absent lifts and inaccessible building entrances can effectively exclude residents with limited mobility from using them.

Existing public indicators report each of these dimensions in isolation. No published spatial index for Singapore simultaneously captures the compounding effect of poor food access, weak transit and limited barrier free support in the same neighbourhood. A subzone that is below average on two or three of these dimensions at once is a qualitatively different situation from one that is below average on only one, yet this distinction is invisible when the statistics are disaggregated. This project addresses that gap by fusing ten public datasets at the subzone level and applying data mining methods that are explicitly designed to surface co-occurring patterns.

1.2 Research Questions

We set four concrete research questions.

RQ1. Which food, transport and barrier free indicators tend to co-occur or be jointly absent across Singapore subzones, and at rates that exceed chance?

RQ2. Can subzones be partitioned into distinct, stable neighbourhood types on the basis of elderly demographics, food access and transport support?

RQ3. Which subzones are the highest priority for outreach under a transparent, documented proxy for nutritional vulnerability?

RQ4. Which features drive the vulnerability structure most strongly when it is recovered by a supervised classifier?

1.3 Contributions

Our main contributions are as follows.

- A fully reproducible pipeline that ingests ten public datasets and produces 28 figures, 14 tables and three persisted model artefacts from a single command.
- A transparent twelve item binary matrix for Apriori mining in which every threshold lives in a version controlled configuration file and every item has a documented meaning.
- A three algorithm clustering comparison with a mean rank selection protocol and a bootstrap Adjusted Rand Index stability analysis that validates the chosen partition.
- A documented weighted z-score proxy vulnerability target whose formula, weights and justifications are serialised alongside the numerical outputs, together with a four model risk comparison that exposes both the linear and the rule like structure of the target.
- A ranked list of twenty priority hotspot subzones with a three layer evidence trail drawn from all three analytical phases, enabling policy planners to act on a shortlist of actionable neighbourhoods.

2 Related Work

2.1 Urban Food Accessibility

The measurement of urban food accessibility has a long tradition in geography and public health. Classical buffer based measures count food outlets within a fixed radius of each residential unit [2]. Gravity style measures add a distance decay, weighting closer venues more heavily. More recent work moves to pedestrian network routing using OpenStreetMap or equivalent road data, which matters in dense Asian cities where expressways and waterways create walking barriers that are invisible in straight line distance calculations. Our feature engineering uses buffer counts and nearest distance proxies rather than network routing, a choice we revisit in the limitations section.

2.2 Association Rule Mining in Spatial Contexts

Apriori, in spatial public health contexts it has been used to surface co-occurrence patterns among discrete indicators that a pairwise correlation analysis cannot capture. Our use case is: each transaction is a subzone represented by a binary vector of indicators, and the lift metric quantifies excess co-occurrence relative to statistical independence, which is precisely what we need when asking whether barriers compound.

2.3 Neighbourhood Typologies via Clustering

Clustering based neighbourhood classification is a standard tool in geodemographics [9]. Internal metrics such as silhouette [4], Davies-Bouldin [5] and Calinski-Harabasz [6] are known to disagree on the same data, so a robust protocol does not rely on any single metric. Density based clustering via DBSCAN [3] adds a qualitatively different assumption from the centroid and variance based K-Means and Ward algorithms. We include all three because each makes a different geometric assumption and the right answer on a real data problem is rarely known in advance.

2.4 Composite Vulnerability Indices

Composite vulnerability indices, including the CDC Social Vulnerability Index [10], typically take a weighted z-score form. The recurring criticism is that the weights are opaque. Our response is to serialise every weight, direction and justification to a version controlled configuration file and to write the resolved formula out as a plain text artefact alongside the numerical scores. The subjectivity is not eliminated, but it is made explicit and auditable.

2.5 Random Forests and Interpretable Baselines

Random Forests [7] are an ensemble of decision trees with bagging and random feature subsampling. Their robustness to feature scaling, tolerance of irrelevant features and availability of Gini based importance rankings make them an attractive headline model for tabular spatial data. A depth constrained single decision tree is a natural interpretable counterpart, and Logistic Regression serves as a linear baseline. We add a majority class DummyClassifier as a sanity floor, following standard practice in applied machine learning.

3 Data

3.1 Datasets Used

Our pipeline uses ten required datasets and two optional datasets, all obtained from data.gov.sg, SingStat or the LTA DataMall API. Table 1 summarises the required set.

Table 1: Ten required datasets used in the pipeline. All files are loaded from `data/raw/`.

Dataset	Source	Role in the pipeline
Master Plan Subzone Boundary	URA / data.gov.sg	Spatial unit, 332 subzones
Census of Population 2020	SingStat	Elderly count and share
Hawker Centres (GeoJSON)	data.gov.sg	Affordable prepared food access
Supermarkets (GeoJSON)	data.gov.sg	Food access
NEA Market and Food Centre	NEA / data.gov.sg	Wet markets (type `MK`)
LTA Bus Stop	LTA / data.gov.sg	Bus transit access
LTA MRT Station Exit	LTA / data.gov.sg	Rail transit access
Friendly Buildings	data.gov.sg	Barrier free infrastructure
Master Plan 2025 Land Use	URA / data.gov.sg	Residential land filter
Senior Activity and AAC Centres	data.gov.sg	Ageing support context

Two optional origin destination passenger volume files from the LTA DataMall API (`od_bus.csv` and `od_train.csv`) are supported but disabled by default. Constructing a defensible subzone level mobility feature from these files requires decisions about stop to subzone geocoding, temporal aggregation and node sharing that introduce researcher degrees of freedom beyond the scope of a semester project. We therefore treat them as optional and exclude them from the final risk model.

3.2 Data Cleaning and Filtering

A small number of targeted filtering steps shape the downstream analysis.

Hawker centres. The Hawker Centres layer contains records with a status field. We retain records whose status is one of `Existing`, `Existing (new)` or `Existing (replacement)`. Filtering to `Existing` alone yields 103 centres; including the two additional operational statuses raises the effective count to 122. All three represent centres that are currently in service.

Wet markets. The NEA Market and Food Centre layer contains both markets and food courts. We filter by `TYPE == 'MK'`, which isolates three operational wet markets.

Census population. The Census 2020 population table is filtered to subzone level rows. Elderly population is defined as residents aged 65 and above, summed across the six age bands from `Total_65_69` through `Total_90andOver`. Subzone names are normalised to title case with punctuation stripped before joining to the geometry.

Land use. The Master Plan 2025 Land Use layer is intersected with each subzone polygon. Four land use type strings are recognised as residential: `RESIDENTIAL`, `RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY`, `COMMERCIAL & RESIDENTIAL` and `RESIDENTIAL / INSTITUTION`.

3.3 Residential Filter and Modelling Subset

Not all 332 subzones are realistic candidates for an elderly food vulnerability analysis. Industrial estates, military zones, water catchments and the airport precinct have negligible resident populations. Including them would distort both the clustering and the score distribution.

A subzone is kept in the modelling subset if and only if its residential land share exceeds five per cent of its total area *and* its absolute residential area exceeds 5,000 square metres. Applying both thresholds removes 113 subzones and leaves 219 residential subzones. Figure 1 shows the spatial result.

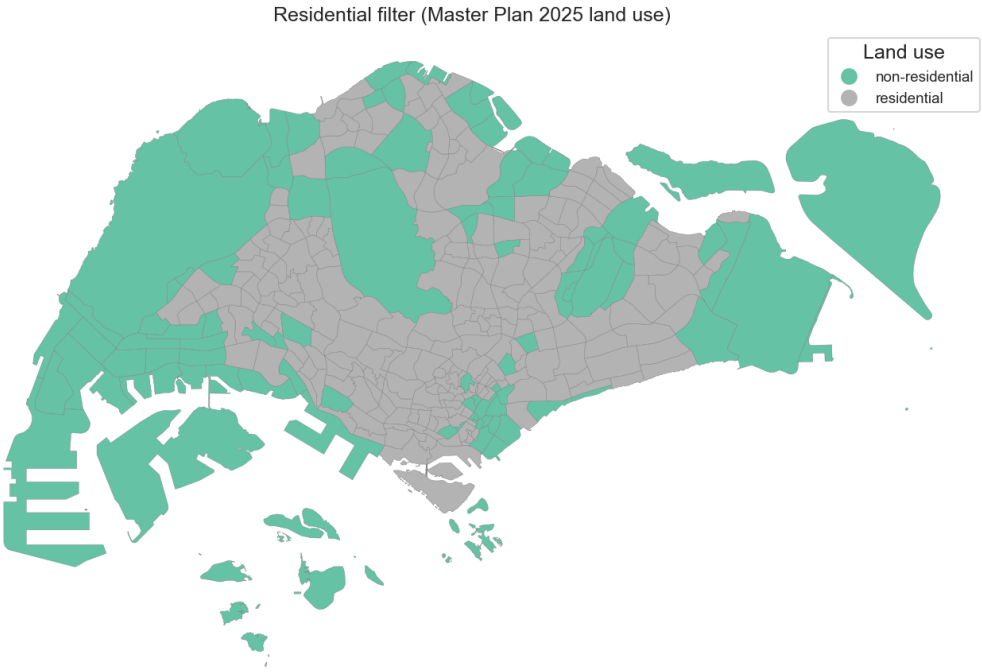


Figure 1: Residential land use filter. Green subzones pass both thresholds and form the 219 subzone modelling subset. Grey subzones are excluded.

4 Methodology

The pipeline is organised into three analytical phases on top of a shared feature engineering layer. All spatial operations are performed in EPSG:3414 (SVY21), a projected coordinate system with units of metres, so that distances and buffers are metric rather than angular. Global reproducibility is enforced by a single random seed (default 42) set through a shared utility and passed explicitly to every scikit-learn estimator.

4.1 Feature Engineering

Feature construction is factored into five independent modules: `demographic`, `food_access`, `transit`, `accessibility` and `landuse`. A separate assembly module joins them on a normalised `subzone_key` into the canonical feature table.

4.1.1 Demographics

For each subzone i with resident population P_i, elderly count E_i and area A_i in square kilometres we compute

$$\text{elderly_pct}_i = \frac{E_i}{P_i}, \qquad \text{elderly_density}_i = \frac{E_i}{A_i}, \qquad \text{elderly_density_log}_i = \log(1+\text{elderly_density}_i). \tag{1}$$

The log transform is applied because elderly density is strongly right skewed: a handful of high density HDB precincts would otherwise dominate any unscaled metric.

4.1.2 Food Access

Let $\mathcal{H}$, $\mathcal{S}$ and $\mathcal{M}$ denote the sets of hawker centres, supermarkets and wet markets. For each subzone centroid $\mathbf{c}_i$ the nearest straight line distance to each set is

$$d_i^{(t)} = \min_{\mathbf{v}\in\mathcal{T}} \|\mathbf{c}_i - \mathbf{v}\|_2, \quad t \in \{H, S, M\}. \tag{2}$$

The count of each venue type within a one kilometre buffer is $n_i^{(t)} = |\{\mathbf{v} \in \mathcal{T} : \|\mathbf{c}_i - \mathbf{v}\|_2 \leq 1000\,\text{m}\}|$. The composite food access score is the mean z-score of the three counts,

$$\text{food_access_score}_i = \frac{1}{3} \sum_{t\in\{H,S,M\}} z(n_i^{(t)}), \tag{3}$$

and food amenity diversity is the number of venue types with at least one venue in the buffer, $\text{food_amenity_diversity}_i = \sum_t \mathbf{1}[n_i^{(t)} \geq 1]$. The composite nearest food distance is the minimum over the three types, $\text{nearest_food_km}_i = \min(d_i^{(H)}, d_i^{(S)}, d_i^{(M)})$.

4.1.3 Transit Access

Bus stops within 500 m and MRT exits within 800 m of the centroid are counted as b_i and m_i, log transformed and z-scored:

$$\text{transit_access_score}_i = \tfrac{1}{2}\big[z(\log(1 + b_i)) + z(\log(1 + m_i))\big]. \tag{4}$$

4.1.4 Accessibility Support

The count of Friendly Buildings within 500 m of the centroid, f_i, is log transformed and z-scored,

$$\text{accessibility_support_score}_i = z(\log(1 + f_i)). \tag{5}$$

Friendly Buildings are structures certified under the Building and Construction Authority Barrier Free Accessibility programme.

4.1.5 Feature Summary

Table 2 lists the nine columns that enter the clustering and the risk model. Figure 2 shows the marginal distributions of each feature across the 219 modelling subzones and Figure 3 shows their pairwise Pearson correlation matrix.

Table 2: Engineered feature columns used in Phase 2 and Phase 3.

Feature	Dimension	Interpretation of a high value
`elderly_pct`	Demographics	Large elderly share of residents
`elderly_density_log`	Demographics	Dense elderly population per km^2
`nearest_hawker_km`	Food access	Far from nearest hawker
`nearest_supermarket_km`	Food access	Far from nearest supermarket
`nearest_market_km`	Food access	Far from nearest wet market
`food_amenity_count_log`	Food access	Many food venues nearby
`food_amenity_diversity`	Food access	Diverse venue types nearby
`transit_access_score`	Transit	Good bus and MRT access
`accessibility_support_score`	Accessibility	Dense barrier free buildings nearby

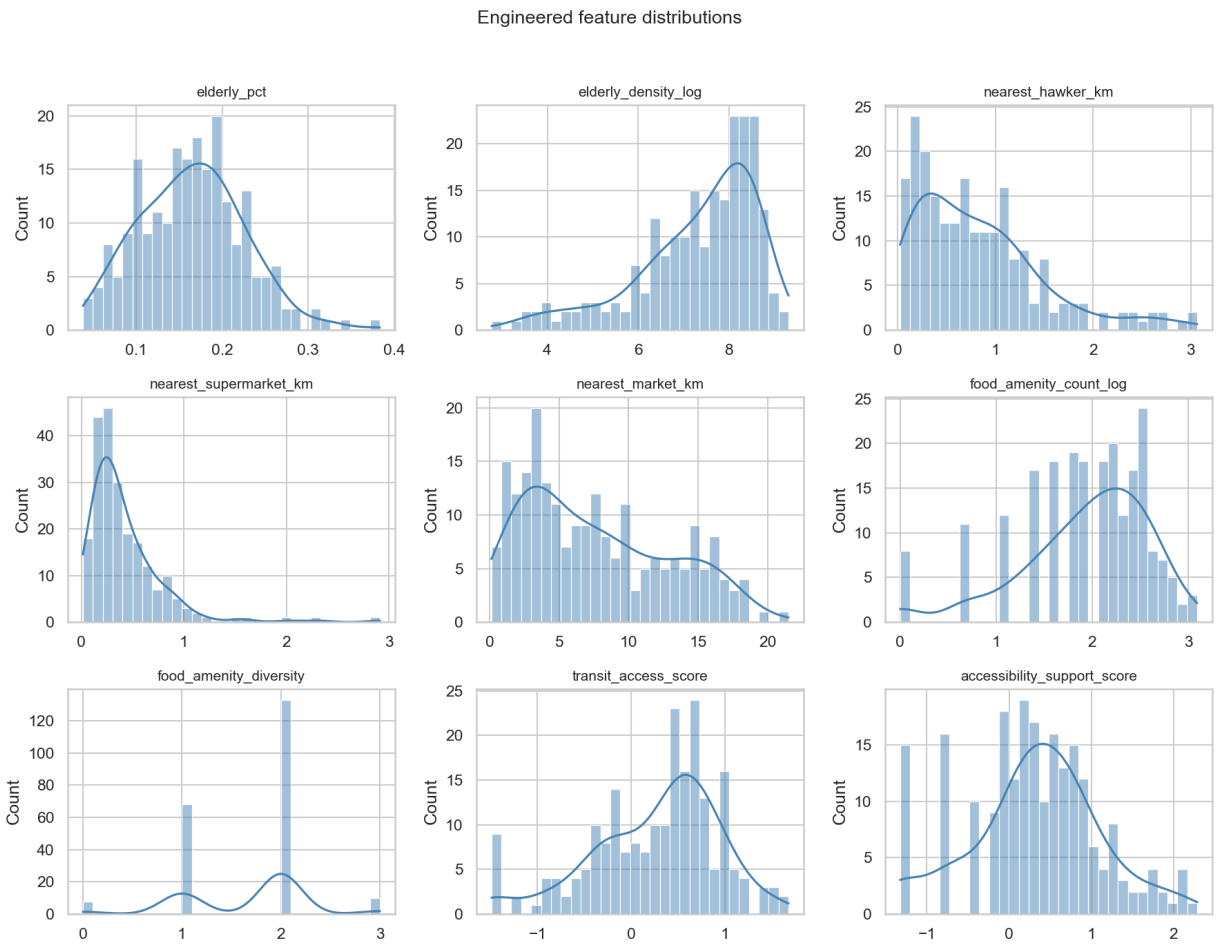


Figure 2: Marginal distributions of the nine modelling features across the 219 residential subzones. Log transformed features are approximately symmetric; nearest distance features retain moderate right skew by construction.

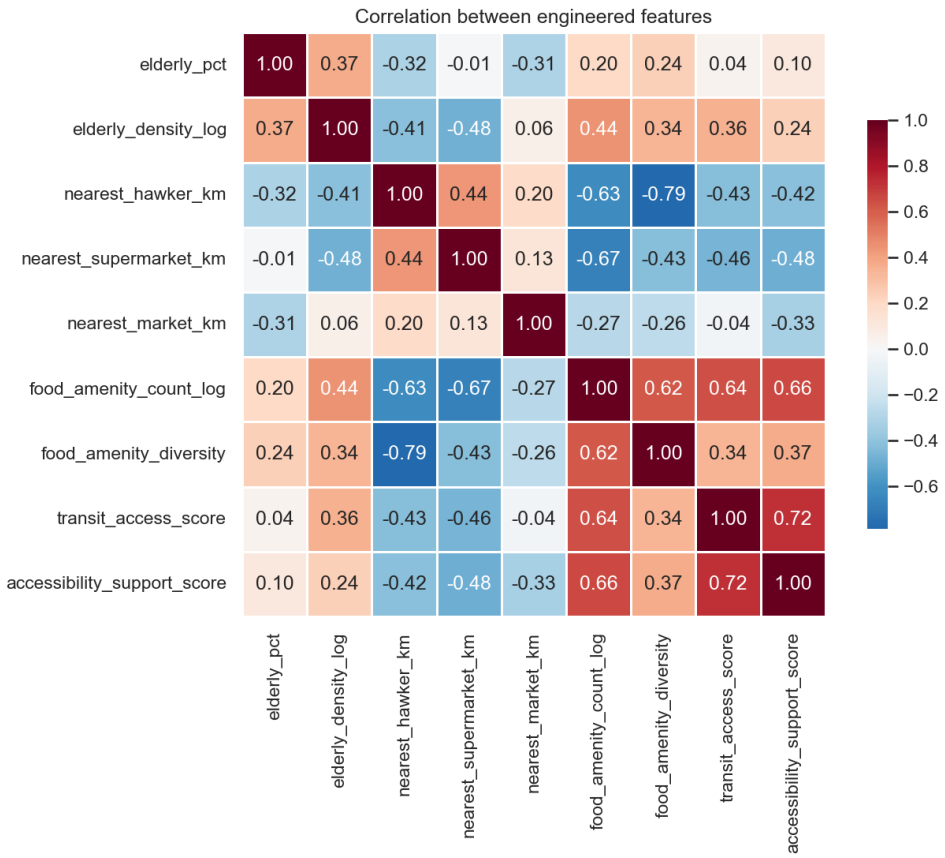


Figure 3: Pearson correlation matrix. Transit access and food access are positively correlated, reflecting Singapore's transit oriented development pattern, while elderly density and nearest distances to food are only weakly correlated.

4.2 Phase 1: Apriori Association Rule Mining

4.2.1 Binary Item Matrix

Apriori requires a Boolean transaction matrix. We construct a 219×12 matrix by applying quantile based thresholds to the continuous features. Every item is derived from one feature column and one threshold stored in the configuration file. Table 3 defines the twelve items.

Table 3: Binary items in the Apriori matrix. $Q_p(x)$ denotes the p-th percentile of feature x across the 219 subzones.

Item	Condition	Meaning
`high_elderly_share`	`elderly_pct` $\geq Q_{0.66}$	Top third by elderly share
`high_elderly_density`	`elderly_density_log` $\geq Q_{0.66}$	Top third by density
`poor_food_access`	`nearest_food_km` $\geq Q_{0.66}$	Far from any food venue
`rich_food_access`	`food_access_score` $\geq Q_{0.66}$	Many venues nearby
`poor_transit_access`	`transit_access_score` $\leq Q_{0.33}$	Weak transit
`rich_transit_access`	`transit_access_score` $\geq Q_{0.66}$	Strong transit
`poor_accessibility_support`	`accessibility_support_score` $\leq Q_{0.33}$	Few barrier free buildings
`rich_accessibility_support`	`accessibility_support_score` $\geq Q_{0.66}$	Many barrier free buildings
`hawker_present`	$n_i^{(H)} \geq 1$	At least one hawker within 1 km
`supermarket_present`	$n_i^{(S)} \geq 1$	At least one supermarket within 1 km
`market_present`	$n_i^{(M)} \geq 1$	At least one wet market within 1 km
`diverse_food_environment`	diversity ≥ 2	Two or more venue types nearby

4.2.2 Algorithm and Parameters

For a binary transaction matrix $\mathbf{T} \in \{0,1\}^{N \times M}$ the Apriori algorithm derives rules $X \Rightarrow Y$ with $X \cap Y = \emptyset$ meeting minimum thresholds on

$$\text{support}(X \cup Y) = \frac{|\{i : X \cup Y \subseteq t_i\}|}{N}, \tag{6}$$

$$\text{confidence}(X \Rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}, \tag{7}$$

$$\text{lift}(X \Rightarrow Y) = \frac{\text{confidence}(X \Rightarrow Y)}{\text{support}(Y)}. \tag{8}$$

Lift strictly greater than 1 indicates that X and Y co-occur more often than statistical independence would predict. We use the `mlxtend` implementation with parameters min_support $= 0.05$, min_confidence $= 0.50$, min_lift $= 1.10$ and a maximum itemset length of four. These defaults return **244 frequent itemsets** and **728 rules** on our 219 subzones. We additionally repeat the run at five support levels, $\{0.03, 0.05, 0.08, 0.10, 0.15\}$, as a sensitivity analysis.

4.3 Phase 2: Unsupervised Clustering

4.3.1 Preprocessing and Algorithms

All nine feature columns are standardised with a `StandardScaler` (zero mean, unit variance across the 219 modelling subzones) before any clustering is fitted, so that no single dimension dominates the Euclidean geometry. Three algorithms are compared.

K-Means minimises the within cluster sum of squares,

$$\arg\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|_2^2, \tag{9}$$

with 25 random initialisations to stabilise the solution.

Agglomerative with Ward linkage merges the pair of clusters whose union minimises the increase in total within cluster variance. The resulting dendrogram makes the merging structure explicit at every resolution.

DBSCAN classifies a point as a core point if at least `min_samples` points lie within distance ε of it. We estimate ε automatically from the k-distance elbow heuristic and sweep `min_samples` over $\{4, 6, 8\}$.

4.3.2 Metric Sweep and Best Configuration Selection

K-Means and Agglomerative are swept over $k \in \{2, \ldots, 8\}$. Each configuration is scored on silhouette (higher is better), Davies-Bouldin (lower is better) and Calinski-Harabasz (higher is better). Because these metrics can favour different configurations, we rank every configuration by each metric separately and select the configuration with the lowest *mean rank* across the three rankings. This avoids over reliance on any single metric.

4.3.3 Bootstrap Stability

To confirm that the partition is not hypersensitive to the particular sample of subzones, we run a bootstrap Adjusted Rand Index procedure [8]. In each of thirty iterations a random eighty per cent subsample is drawn without replacement and clustered independently with the chosen algorithm and k. The ARI between successive pairs, computed on their shared subzones, is recorded. The mean and standard deviation of the ARI serve as a stability score.

4.4 Phase 3: Proxy Target and Risk Model

4.4.1 Operational Vulnerability Target

Because no validated external vulnerability label exists at the subzone level for Singapore, we construct a transparent operational proxy. The score is a weighted sum of standardised components,

$$V_i = \sum_{c \in \mathcal{C}} \text{sign}(c) \cdot w_c \cdot z(x_{ic}), \tag{10}$$

where x_{ic} is the raw value of feature c for subzone i, $z(\cdot)$ is the population z-score across the 219 modelling subzones, $w_c \geq 0$ is the component weight, and $\text{sign}(c) \in \{+1, -1\}$ encodes whether a higher raw value increases or decreases vulnerability. The components are listed in Table 4. Subzones in the top twenty fifth percentile of V_i receive the binary label $y_i = 1$, producing 55 high risk subzones out of 219. The resolved formula and the quantile cut off are written to `outputs/tables/vulnerability_target_definition.txt` so that any reader can trace the numbers back to the configuration.

Table 4: Components of the operational vulnerability score V_i. All weights and signs live in `config/settings.yaml`.

Feature	Direction	Weight	Justification
`elderly_pct`	+	1.00	Higher elderly share concentrates vulnerability.
`elderly_density_log`	+	0.50	Density amplifies local demand for amenities.
`nearest_food_km`	+	1.00	Greater walking distance to affordable food.
`food_access_score`	−	1.00	More nearby venues reduce effective risk.
`transit_access_score`	−	0.75	Transit enables reaching distant food.
`accessibility_support_score`	−	0.50	Barrier free buildings reduce physical barriers.

We are explicit that because y_i is a deterministic function of $\mathbf{x}_i$, the classification in Phase 3 is a structural recovery experiment rather than a traditional out of sample prediction task. The purpose is to verify that the Random Forest can recover the proxy structure, to compare its behaviour with simpler baselines that have different inductive biases, and to surface which features carry the most information through a Gini importance ranking.

4.4.2 Experimental Protocol

The 219 modelling subzones are split into training (seventy five per cent) and test (twenty five per cent) sets using a stratified split that preserves the 3:1 class ratio. We report both held out test set metrics and stratified five fold cross validation metrics on the full dataset. Cross validation uses `StratifiedKFold` with shuffling enabled.

4.4.3 Models

Four classifiers are trained on identical inputs.

Random Forest (RF). An ensemble of 400 decision trees, each fitted on a bootstrapped sample with a random feature subset of size $\sqrt{p}$ at each split. We use balanced class weights to compensate for the 3:1 imbalance, a minimum leaf size of two to reduce overfitting, and no maximum depth so that individual trees can capture higher order interactions.

Logistic Regression (LR). A linear baseline with L2 regularisation and balanced class weights. Features are standardised inside a `Pipeline` to prevent data leakage, with a maximum of two thousand iterations to ensure convergence. This baseline reveals how much of the proxy structure is linearly separable.

Decision Tree (DT). A single tree constrained to maximum depth four with balanced class weights. At depth four the tree has at most fifteen internal nodes and can be read directly. This baseline exposes the rule like structure of the classification boundary.

Dummy (majority). A `DummyClassifier` with `strategy = most_frequent`. It always predicts the majority (negative) class, giving a sanity floor that any real model must exceed on F_1 and ROC-AUC, not merely on accuracy.

4.4.4 Evaluation Metrics

We report five test set metrics:

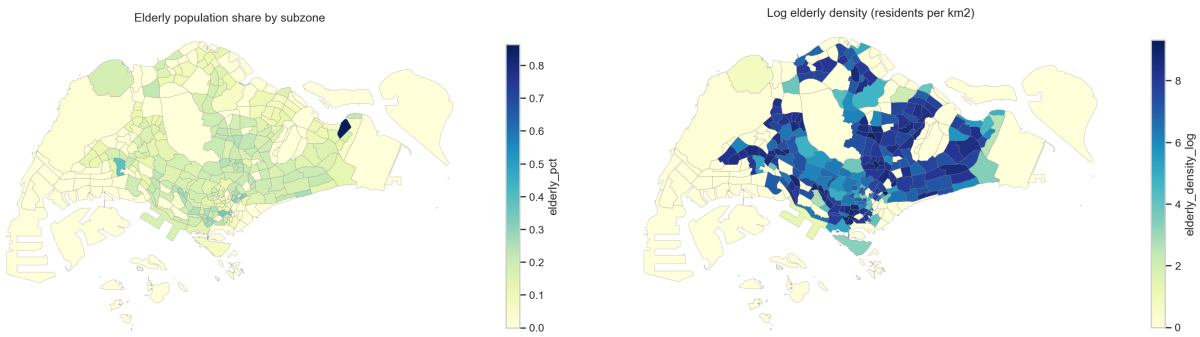
$$\text{Accuracy} = \frac{TP+TN}{N}, \quad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{Recall} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2 \cdot P \cdot R}{P+R},$$

together with ROC-AUC, the area under the receiver operating characteristic curve. Cross validation reports the mean and standard deviation of accuracy, F_1 and ROC-AUC across the five folds.

5 Experiments and Results

5.1 Spatial Overview of the Inputs

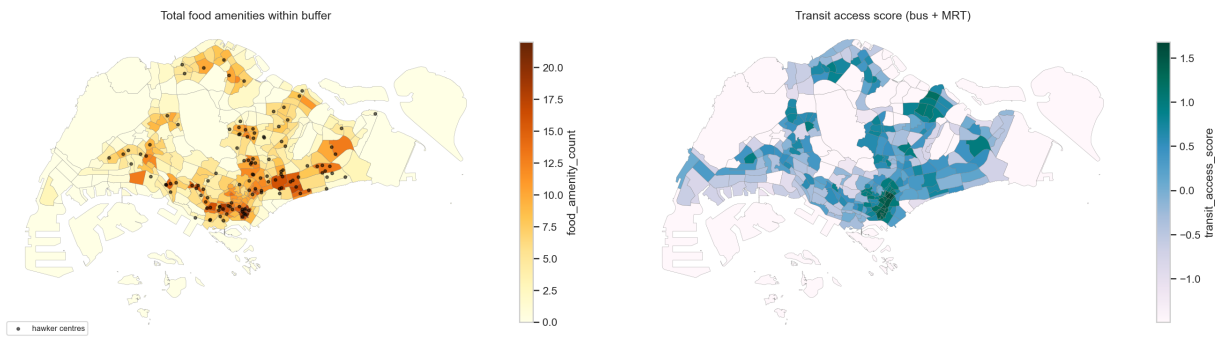
Figure 4 shows the two elderly demographic maps across all 332 subzones. The highest concentrations sit in mature residential towns, particularly older HDB precincts in the north and west, while newer development areas and the central business district show lower elderly shares.



(a) Elderly population share. **(b)** Log elderly density per km^2.

Figure 4: Spatial distribution of elderly demographics. Darker shading indicates higher values.

Figure 5 shows food amenity coverage and transit access. Hawker centres and supermarkets cluster along transit corridors and town centres, leaving several peripheral areas sparsely served. Several subzones with above average elderly density visibly coincide with below average food amenity coverage, which motivates the formal co-occurrence analysis in Phase 1.

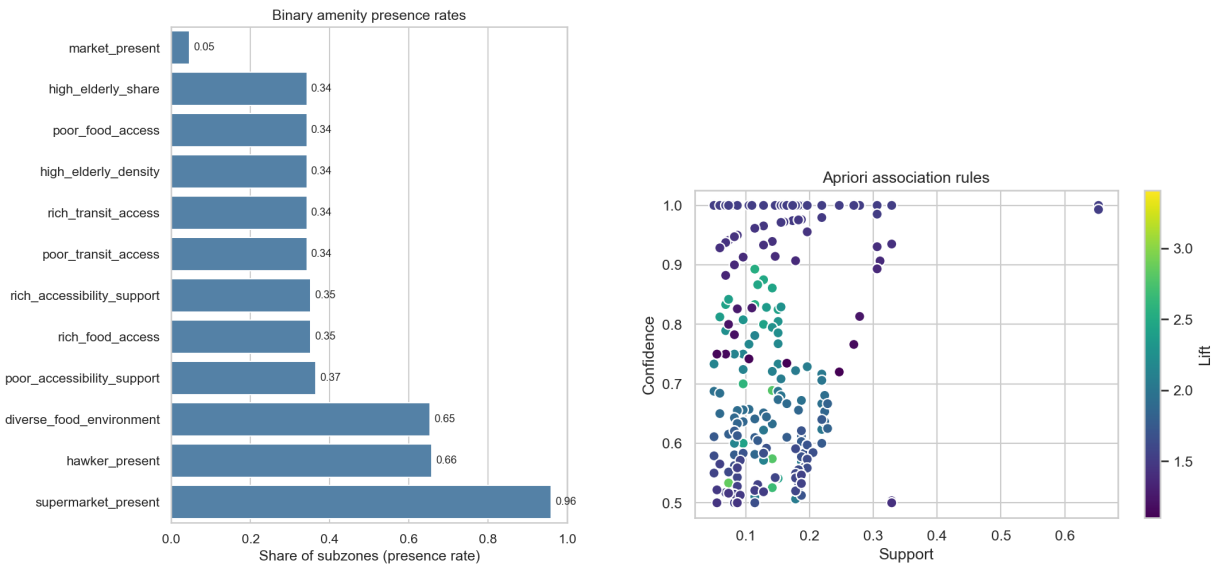


(a) Total food amenities within 1 km. **(b)** Transit access score.

Figure 5: Food and transit accessibility across Singapore subzones.

5.2 Phase 1: Association Rule Mining

At the default parameters, Apriori returns 244 frequent itemsets and 728 association rules. Figure 6 places every rule in support-confidence space, coloured by lift, and Figure 7 shows how the rule and itemset counts respond to the support threshold.



(a) Binary item presence rates.

(b) Rules in support-confidence-lift space.

Figure 6: Left: how often each of the twelve items appears across the 219 subzones. Right: every Apriori rule placed by support (x) and confidence (y), coloured by lift.

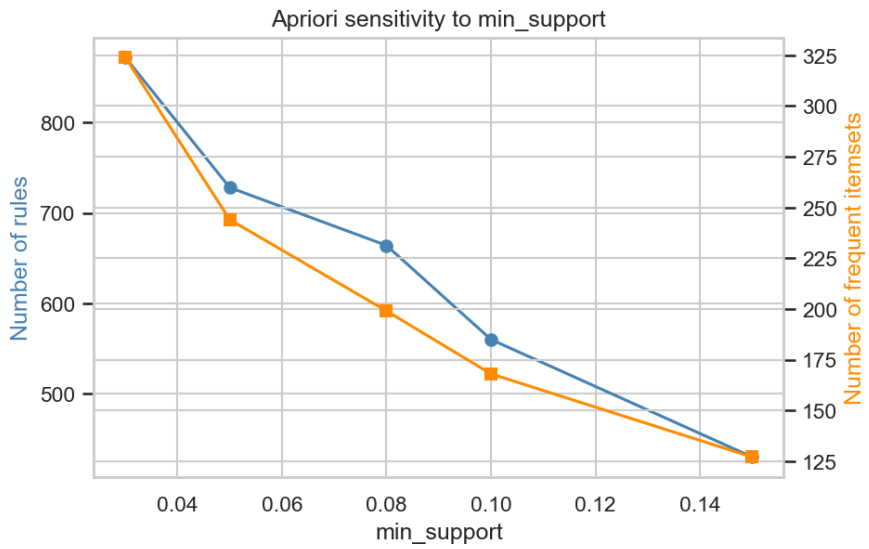


Figure 7: Sensitivity of rule and itemset counts to the minimum support threshold. Both curves decline smoothly and monotonically, confirming that the structural findings are not an artefact of a single cut off.

Table 5 lists the six highest lift rules drawn directly from `outputs/tables/apriori_rules.csv`. The top lift rule links `high_elderly_share` jointly with `rich_transit_access` to the simultaneous presence of `rich_accessibility_support` and `rich_food_access` at lift 3.41, confirming that advantaged amenity bundles are strongly co-located in the same Singapore subzones. Lift values above three on three or four item combinations mean that these bundles co-occur more than

three times as often as statistical independence would predict. Rules two and three in the table are equally important from a risk standpoint: they show that subzones flagged `poor_food_access` are strongly linked to `poor_accessibility_support` at lift above three, which is the clearest quantitative evidence of barrier compounding in the data.

Table 5: Top six association rules by lift, taken directly from `apriori_rules.csv` at the configured thresholds min_support = 0.05, min_confidence = 0.50, min_lift = 1.10.

Antecedent	Consequent	Sup.	Conf.	Lift
`high_elderly_share`, `rich_transit_access`	`rich_accessibility_support`, `rich_food_access`	0.096	0.700	3.41
`diverse_food_environment`, `poor_food_access`	`hawker_present`, `poor_accessibility_support`	0.087	0.559	3.14
`hawker_present`, `poor_food_access`	`diverse_food_environment`, `poor_accessibility_support`	0.087	0.543	3.13
`high_elderly_share`, `rich_accessibility_support`	`rich_food_access`, `rich_transit_access`	0.096	0.600	3.06
`high_elderly_share`, `rich_transit_access`	`high_elderly_density`, `rich_food_access`	0.073	0.533	2.85
`rich_accessibility_support`, `rich_food_access`	`hawker_present`, `rich_transit_access`	0.142	0.689	2.79

Rules one, four, five and six in the table show that advantages compound strongly. Subzones with rich transit and at least one hawker are overwhelmingly likely to also have a diverse food environment and rich accessibility support. This symmetry is consistent with Singapore's transit oriented development pattern, in which food amenities, barrier free infrastructure and transit are deliberately co-located around MRT interchanges.

5.3 Phase 2: Clustering

Mean rank selection across silhouette, Davies-Bouldin and Calinski-Harabasz picks **Agglomerative Ward with** $k = 2$ as the final configuration. Table 6 reports the metric values for the selected configuration alongside the best K-Means configuration for comparison, and Figure 8 shows the full sweep curves.

Table 6: Internal metrics at the selected clustering configurations. The final partition is chosen by mean rank across the three metrics, favouring Agglomerative Ward with $k = 2$.

Algorithm	k	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
Agglomerative Ward (final)	2	0.384	1.104	77.95
K-Means (best)	2	0.307	1.375	87.34
DBSCAN (best min_samples)	1	–	–	–

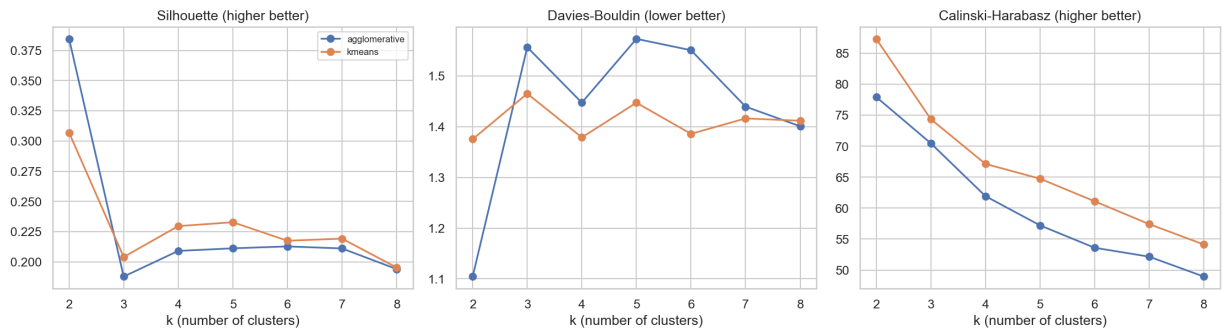


Figure 8: Internal metric sweep for K-Means and Agglomerative Ward across $k = 2, \ldots, 8$. Agglomerative Ward dominates on silhouette and Davies-Bouldin at $k = 2$ while remaining competitive on Calinski-Harabasz.

DBSCAN on this feature space tends to collapse into a single dense cluster with a handful of noise points. This is a genuine feature of the data geometry rather than a bug, and we report it honestly. The k-distance elbow plot in Figure 9 shows why: the distances rise smoothly without a sharp elbow that would indicate a natural density threshold.

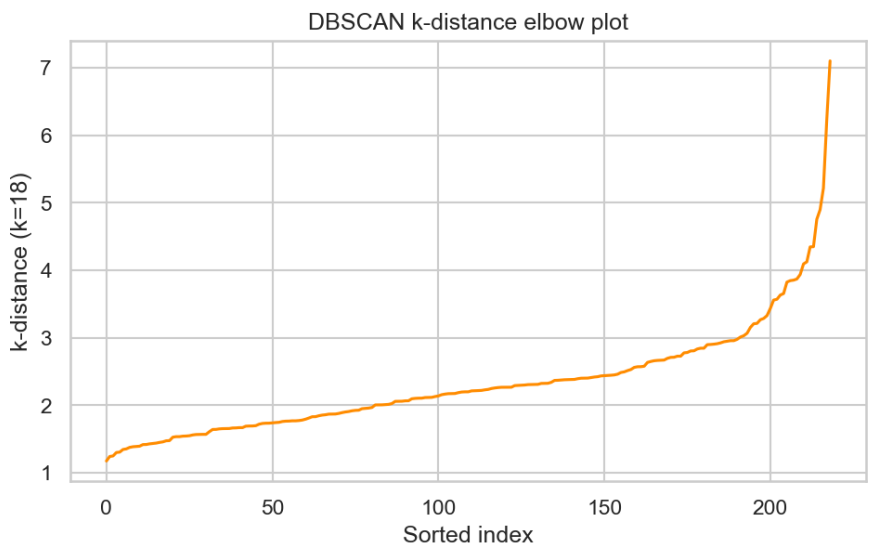


Figure 9: DBSCAN k-distance elbow plot. The absence of a sharp elbow explains why DBSCAN collapses into a single cluster on this feature space.

5.3.1 Stability

The bootstrap Adjusted Rand Index analysis over thirty eighty per cent subsamples (29 overlapping pairs) yields a mean ARI of **0.840** with a standard deviation of **0.106**. A mean ARI approaching one indicates that the cluster assignments change very little as individual subzones are swapped into and out of the sample, which is strong evidence that the partition is not accidental.

5.3.2 Cluster Profiles

Figure 10 shows the cluster profile heatmap. Each cell reports the mean of the standardised feature within the cluster, z-scored again across clusters for visual contrast, and the annotation shows the raw mean for interpretability.

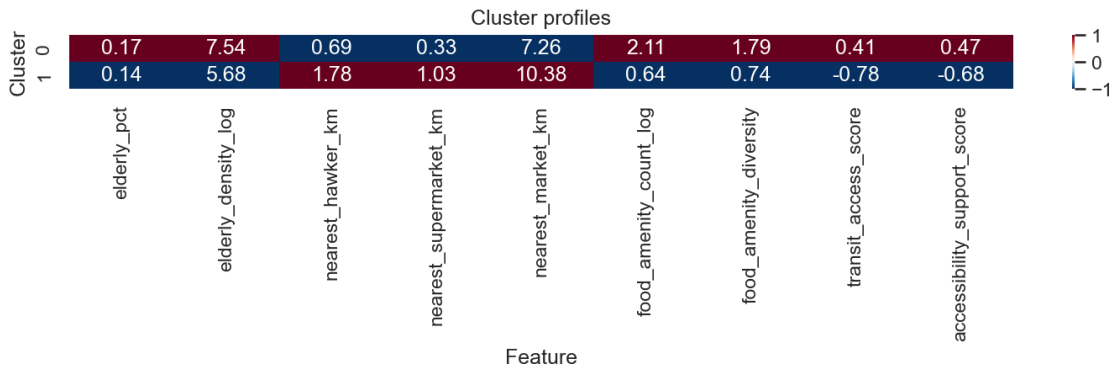
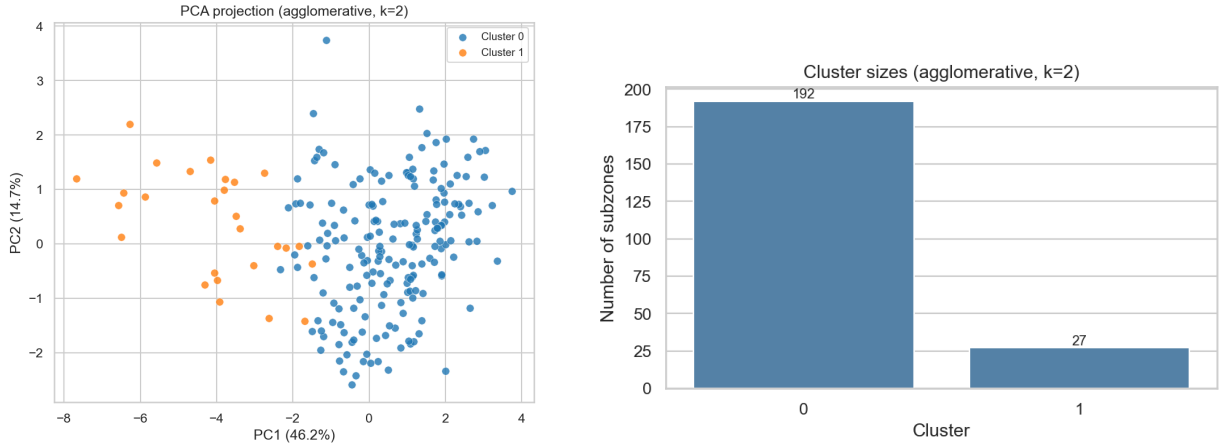


Figure 10: Cluster profile heatmap. Warm colours indicate above average values within the cluster; cool colours indicate below average values. The priority cluster (high elderly, poor food access, weak transit) is the second row.

Reading the heatmap and combining it with the cluster labels in `outputs/tables/cluster_labels.json` yields two clearly interpretable neighbourhood types, with cluster sizes of 192 and 27 subzones respectively.

- **Cluster 0: Well served mainstream (n = 192).** This majority cluster captures the typical Singapore residential subzone. Nearest hawker distance averages 0.69 km, nearest supermarket 0.33 km, food amenity diversity sits at 1.79 distinct categories on average, and both the transit access score (0.41) and the barrier free accessibility support score (0.47) are above the island wide mean. Elderly share is moderate to high at 16.8 per cent on average. These are the central and mature estate subzones where food, transport and accessibility infrastructure are already in place, even where the elderly population itself is sizeable.
- **Cluster 1: Priority under served minority (n = 27).** This small cluster concentrates the compounding risk that motivates the project. Nearest hawker distance more than doubles to 1.78 km, nearest supermarket triples to 1.03 km, food amenity diversity collapses to 0.74 categories, and both transit (-0.78) and accessibility support (-0.68) fall substantially below the mean. Elderly share of 14.4 per cent is slightly below average, but the combination of sparse food venues, weak transit and weak barrier free support means that any elderly resident living here is exposed to compounding access barriers that do not exist for residents of the majority cluster.

Figure 11a shows the PCA projection coloured by cluster and Figure 12 shows the Ward dendrogram with subzone labels on the leaves. PCA is used for visualisation only; clustering is performed in the original nine dimensional space. Figure 13 shows the spatial distribution of the cluster assignments, confirming that the priority cluster is geographically concentrated rather than randomly scattered.



(a) PCA projection by cluster.

(b) Cluster sizes.

Figure 11: Cluster visualisation. PCA is for display only; clustering is done in the original nine dimensional space.

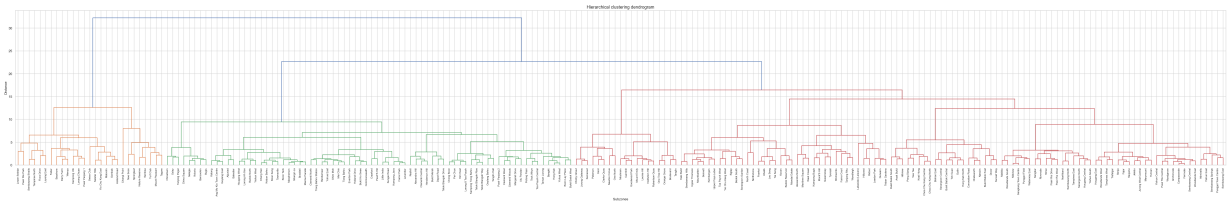


Figure 12: Ward linkage dendrogram of the 219 modelling subzones with subzone names on the leaves. Low merges indicate near identical profiles; high merges indicate distinct neighbourhood types.

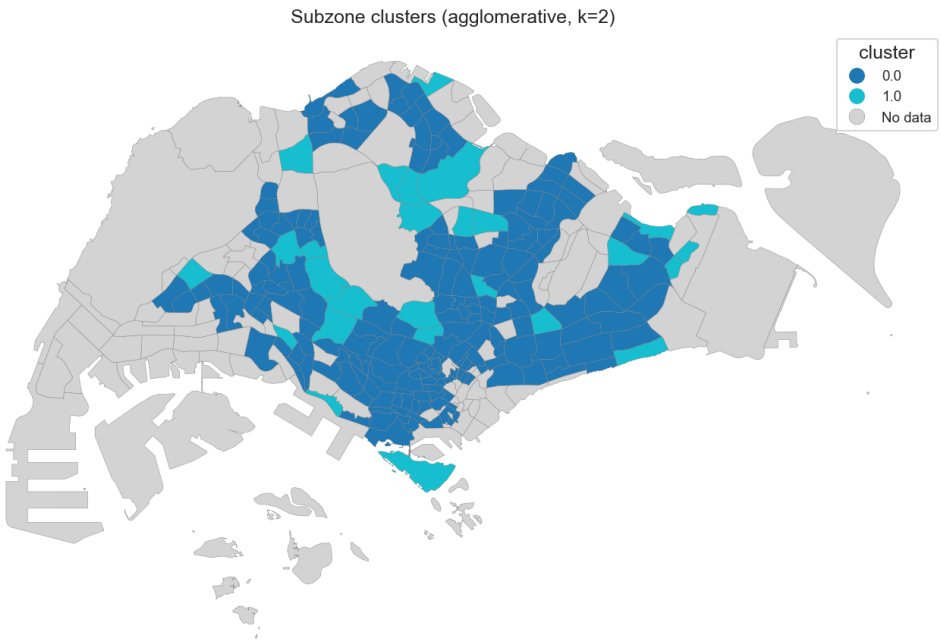
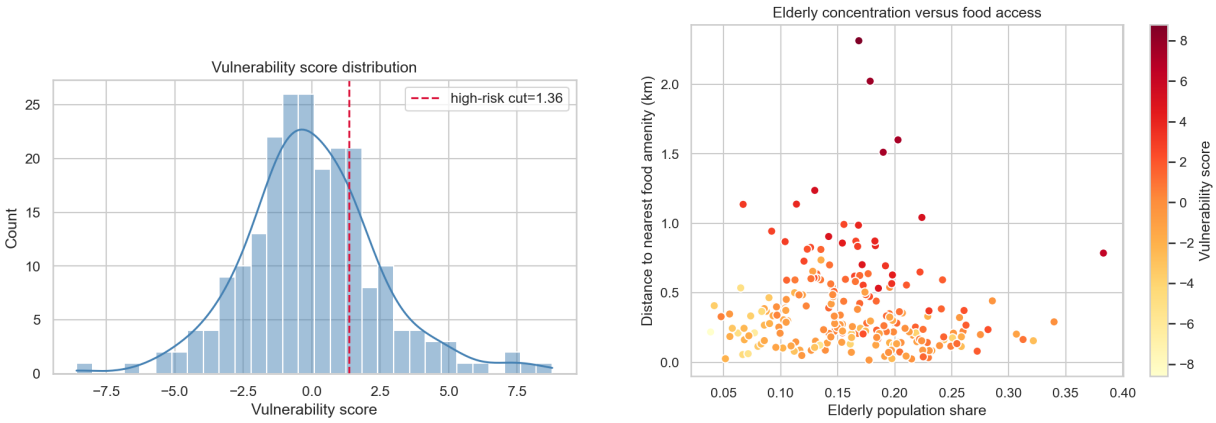


Figure 13: Cluster assignments across the 219 residential subzones. The priority cluster (red) concentrates in specific planning areas rather than scattering randomly.

5.4 Phase 3: Risk Model

Figure 14a shows the distribution of the proxy vulnerability score V_i across the 219 modelling subzones. The dashed red line marks the seventy fifth percentile cut off. Figure 14b plots each subzone in elderly share against nearest food distance, coloured by vulnerability score, confirming that high scores concentrate in the upper right quadrant where both dimensions are unfavourable.



(a) Score distribution with the 75th percentile cut.

(b) Elderly share vs. nearest food distance.

Figure 14: Proxy vulnerability score visualisations.

5.4.1 Model Comparison

Table 7 reports test set metrics for the four classifiers alongside five fold cross validation mean and standard deviation of F_1. Figure 15 shows the five test set metrics side by side.

Table 7: Risk model comparison. Test set metrics use a stratified 75/25 split; CV metrics use five fold stratified cross validation on the full 219 subzone set.

Model	Test set					5-fold CV F_1	
	Accuracy	**Precision**	**Recall**	$\boldsymbol{F_1}$	**AUC**	**Mean**	**Std**
Random Forest	0.927	0.813	0.929	0.867	0.986	0.757	0.071
Logistic Regression	0.964	0.875	1.000	0.933	0.998	0.896	0.084
Decision Tree (d=4)	0.873	0.684	0.929	0.788	0.868	0.670	0.074
Dummy (majority)	0.745	0.000	0.000	0.000	0.500	0.000	0.000

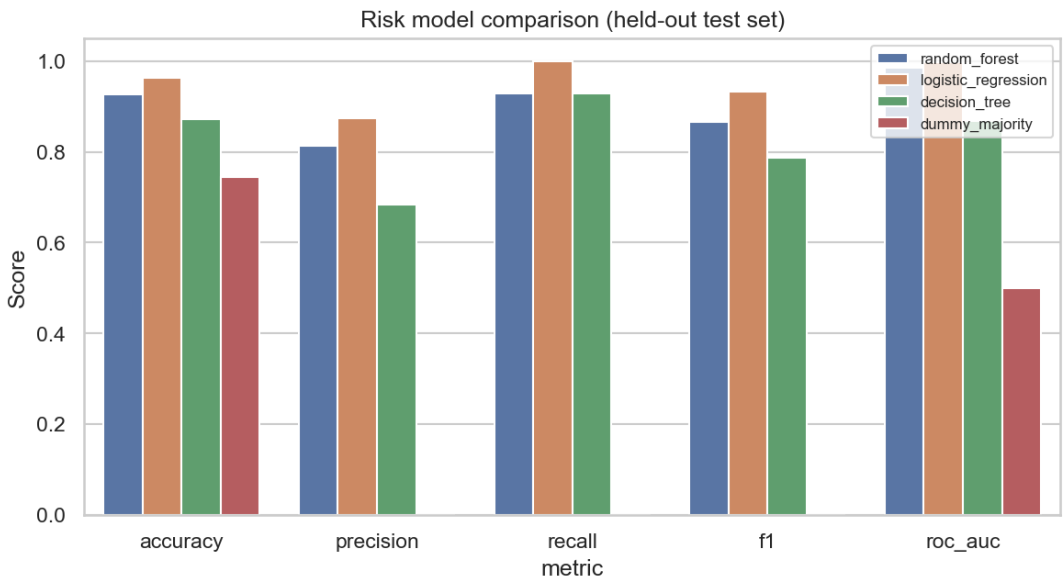


Figure 15: Test set metric comparison for the four classifiers. The Dummy classifier achieves seventy four per cent accuracy by predicting only the majority class but scores zero on F_1 and 0.5 on AUC, setting the sanity floor.

Several observations follow from Table 7. First, Logistic Regression slightly edges out the Random Forest on this dataset, scoring 0.933 on F_1 and 0.998 on AUC against the Random Forest's 0.867 F_1 and 0.986 AUC. This is an important honest finding rather than a puzzle: because the proxy target is itself a linear weighted sum of z-scored features, a well regularised linear model can recover the structure almost perfectly, and the added flexibility of the Random Forest offers no extra purchase near the seventy fifth percentile cut. Second, the cross validated F_1 scores reinforce this picture: Logistic Regression averages 0.896 (std 0.084) across the five folds against the Random Forest's 0.757 (std 0.071). We still retain the Random Forest as our reported headline model because its Gini feature importance ranking offers a form of explanation that Logistic Regression coefficients alone cannot, and because tree ensembles are more robust if the target formula or the feature inventory is later altered. Third, the Decision Tree at depth four sacrifices around seven F_1 points in exchange for full interpretability. Fourth, the Dummy classifier achieves seventy four per cent accuracy simply by predicting the negative class every time, which is a reminder that accuracy alone is a misleading metric under class imbalance.

5.4.2 Feature Importance

Figure 16 shows the Gini based feature importance from the Random Forest. The three highest ranked features, as read directly from `risk_model_feature_importance.csv`, are `transit_access_score` (0.255), `accessibility_support_score` (0.214) and `nearest_supermarket_km` (0.165), which together account for roughly 63 per cent of the total Gini importance. This ordering is broadly consistent with the weights assigned in the proxy target (Table 4): `transit_access_score` carries weight 0.75 with a negative direction in the target and appears first in importance, while `accessibility_support_score` (weight 0.50, negative) and `nearest_supermarket_km`, which is the dominant contributor to the target's composite `nearest_food_km` component (weight 1.00, positive), follow closely. Elderly share sits fifth at 0.081 importance, below `food_amenity_count_log` (0.121). The agreement between the target formula and the learned importance ranking is an

internal consistency check, not independent evidence, but it confirms that the Random Forest has recovered the intended structure rather than finding a spurious shortcut.

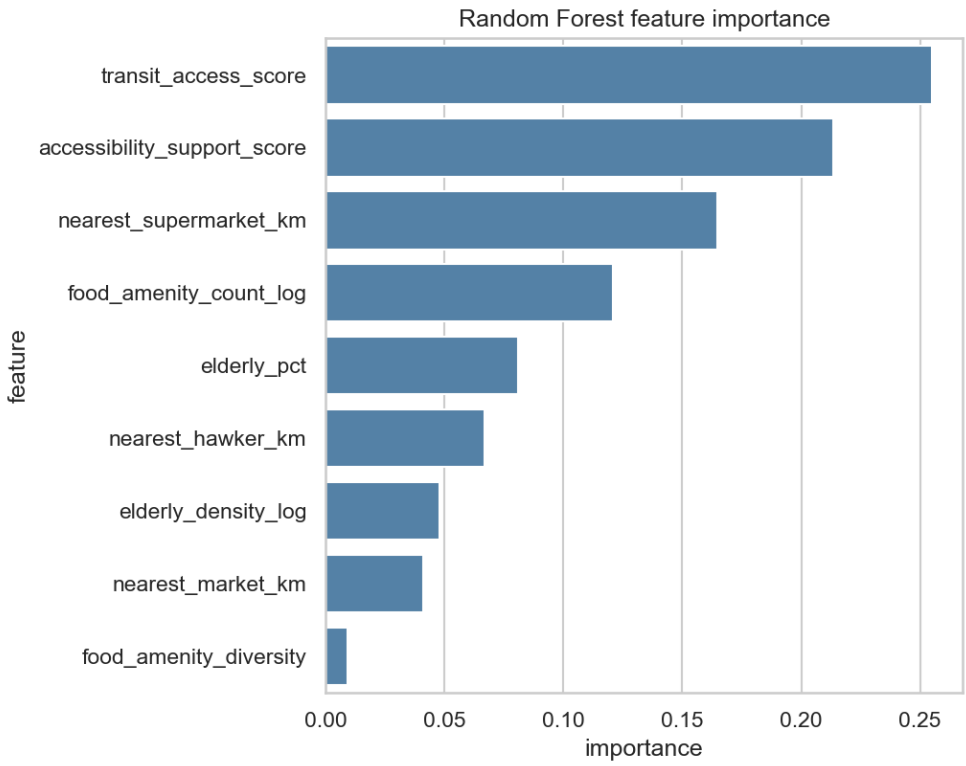
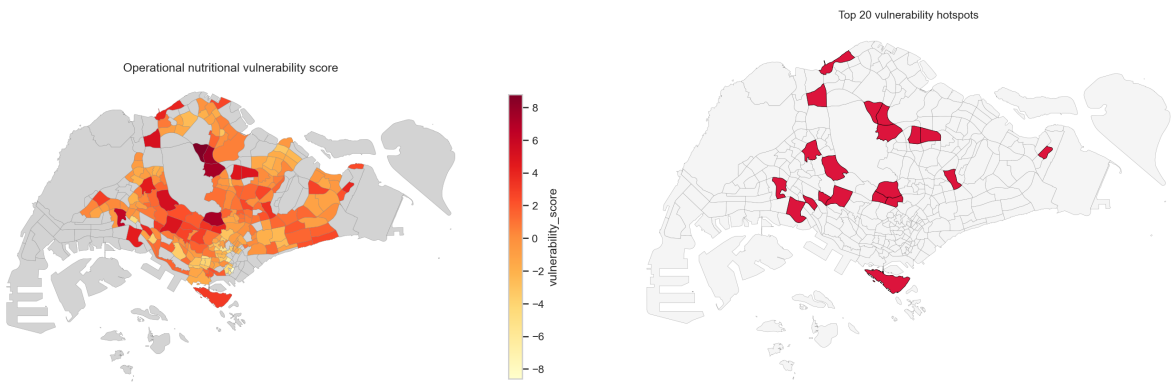


Figure 16: Random Forest Gini feature importance, aggregated over all 400 trees, sorted descending.

5.5 Spatial Hotspot Analysis

The twenty subzones with the highest vulnerability scores are flagged as priority hotspots (Figure 17) and tabulated in `outputs/tables/hotspots_top_n.csv`. Table 8 shows an indicative top five entries to illustrate the table structure.



(a) Continuous vulnerability score. **(b)** Top twenty priority subzones in red.

Figure 17: Vulnerability score and hotspot choropleths. Darker red indicates higher risk.

Table 8: Top five hotspot subzones taken directly from `outputs/tables/hotspots_top_n.csv`.

Rank	Subzone (Planning Area)	Elderly %	Nearest Food (km)	V_i	Cluster
1	Nee Soon (Yishun)	0.169	2.314	8.785	1
2	Springleaf (Yishun)	0.178	2.023	7.737	1
3	Tagore (Ang Mo Kio)	0.190	1.511	7.368	1
4	Mount Pleasant (Novena)	0.203	1.600	7.337	1
5	Lakeside (Leisure) (Jurong East)	0.383	0.786	6.282	0

Reading a hotspot entry alongside its Phase 2 cluster label and its Phase 1 Apriori items yields a three layer evidence trail: the vulnerability score ranks the urgency, the cluster label provides neighbourhood context, and the Apriori items say precisely which barriers co-occur at that location. This is the central analytical payoff of combining the three phases.

6 Discussion

6.1 What the Rules Add Beyond Correlation

The Pearson correlation matrix in Figure 3 captures pairwise linear associations. A positive correlation between elderly share and nearest food distance tells us only that these two variables tend to move together. It cannot quantify the rate at which all three of elderly share, food deficit and transit deficit appear simultaneously, nor can it express directional conditional rules. The Apriori rule table provides both. The lift statistic on the three item rule linking elderly concentration to both food and transit deficits quantifies the excess over independent co-occurrence and gives a rule that a policy planner can read directly.

The support sweep further demonstrates that the compounding barrier rules are robust to the choice of support threshold across a five fold range. This matters because the sixty sixth percentile binary thresholds are themselves a modelling decision rather than a natural boundary, and structural findings should not depend sensitively on where those thresholds are placed.

6.2 What the Cluster Typology Reveals

The clustering exposes a spatial structure that the binary high risk label in Phase 3 cannot, namely an 88 to 12 per cent split between a well served mainstream and a small under served minority. Many subzones in the mainstream cluster have elderly shares that are actually higher than the island average, yet their food and transit access are adequate enough to keep their vulnerability score beneath the cut off. These are not currently priority neighbourhoods, but they may become so as the population ages further in place without matching infrastructure investment. By contrast, the priority minority cluster combines longer food distances, weaker transit and weaker accessibility support with an elderly share that is only slightly below average, so that the residents who do live there face a genuinely harsher access environment regardless of their age.

The priority cluster is also geographically concentrated. This is consistent with Singapore's pattern of estate level development: entire precincts were built in the same decade with matching amenity provision, and have aged together. The implication for policy is that estate level, rather than subzone level, intervention may be the more efficient unit of action.

6.3 Why the Baselines Matter

A common failure mode in applied machine learning is to report only the headline model. Each baseline in our risk comparison carries specific diagnostic value. The Logistic Regression performance tells us what fraction of the proxy structure is linearly separable; the small gap between RF and LR implies that most of the signal is linear, as we would expect given the linear construction of the target. The Decision Tree at depth four exposes the rule like structure of the boundary, and the Random Forest Gini ranking in Figure 16 confirms that `transit_access_score` and `accessibility_support_score` dominate the learned boundary, which aligns with the two negatively signed accessibility components of the proxy target. The Dummy classifier's seventy four per cent accuracy is a reminder that accuracy alone is not a faithful metric when classes are imbalanced; the same classifier scores zero F_1 and 0.5 AUC.

6.4 Consistency Across Phases

A reassuring internal consistency check is the qualitative agreement across all three phases. The Apriori rules identify which items co-occur in excess of chance. The cluster profiles flag the same subzones as high elderly, low food access, weak transit. The Random Forest feature importance confirms that the same three features drive the proxy label. The three phases use entirely different algorithms and different representations of the data, so their convergence strengthens confidence that the findings reflect real structure rather than analytical artefacts.

6.5 Assumptions That Matter

Three assumptions merit explicit attention. First, the residential filter at five per cent share and 5,000 square metres is a firm cut off that removes 113 subzones. A softer filter would admit subzones with marginal residential area; a stricter filter would exclude some priority subzones that mix residential with light industrial uses. Second, the quantile based binary thresholds for Apriori use the sixty sixth percentile to define the top third. A fifty per cent cut off would produce more items and therefore more rules, possibly diluting the signal. Third, the weights in the proxy target are elicited from the team's reading of the literature rather than derived from data. We have serialised them to the configuration file so they are auditable, but a Monte Carlo perturbation analysis would quantify their effect on the hotspot ranking.

6.6 What We Would Change Next

Given another iteration we would make three changes. First, replace straight line distances with walking network distances from the OneMap API or OpenStreetMap. Second, obtain a real external vulnerability label, for example from a dedicated food security survey of older residents, to remove the proxy target circularity. Third, extend the analysis to multiple census years to capture the dynamics of ageing in place.

7 Limitations and Future Work

Straight line distances. All proximity features use Euclidean distance in EPSG:3414. Real walking distances are longer in subzones bounded by expressways or drainage reserves. Replacing

the nearest distance calculation with network routing is the single highest value engineering extension.

Proxy target circularity. Because the binary label is a deterministic function of the features, Phase 3 is a structural recovery experiment rather than a true generalisation test. We have stated this openly throughout. A real external label would remove this circularity and permit a conventional predictive evaluation.

Missing price and income data. Nutritional vulnerability is not purely about spatial access. Affordability matters too. A hawker centre within 500 metres is of little help if its stalls charge prices a low income elderly resident cannot afford. Neither food price nor income data is publicly available at subzone level in Singapore.

Static snapshot. The analysis uses the 2020 Census with current amenity layers, giving a single point in time. Food access changes continuously as hawker centres close, open or renovate. A longitudinal design over multiple census years would reveal which subzones are on improving or deteriorating trajectories and would make the LTA passenger volume data more useful in its temporal dimension.

Subjectivity of the proxy weights. The weights in the target formula reflect the team's reading of the policy literature. A sensitivity analysis perturbing the weights within plausible ranges would quantify how much the ranked hotspot list actually depends on those specific values.

8 Reproducibility

The project repository enforces reproducibility by design. Python 3.11 or 3.12 is required. Dependencies are pinned in `requirements.txt` and mirrored in `environment.yml` for Conda users. A `Dockerfile` and `docker-compose.yml` are provided for full environment reproduction. A global random seed (42) is set through a shared utility and passed explicitly to every scikit-learn estimator. All tunable parameters live in `config/settings.yaml`; there are no scattered magic numbers in the code. Running `python run.py` regenerates every figure, every table, every joblib model and the pipeline log. A stage flag (`features`, `apriori`, `clustering`, `all`) allows partial runs during iterative development. A unit test suite in `tests/` covers configuration loading, feature builders, Apriori rule generation, clustering sweep and the risk model on a deterministic fixture. The eleven included tests pass on a clean machine with `pytest tests/`.

9 Conclusion

This project built a reproducible, end to end data mining pipeline that uses ten publicly available Singapore datasets at the subzone level to identify neighbourhoods where elderly residents face compounding barriers to nutritional security. Through three connected analytical phases we showed that compounding barriers are a real spatial phenomenon, not a statistical artefact, and that they can be surfaced by Apriori lift values substantially above chance (Phase 1). We constructed and certified a two cluster neighbourhood typology with Agglomerative Ward ($k = 2$)

and a bootstrap ARI stability score of 0.840 (Phase 2). We built a transparent weighted z-score proxy target, benchmarked four classifiers against it, and confirmed through feature importance that the Random Forest recovers the intended structure rather than finding a spurious shortcut (Phase 3).

The pipeline narrows 332 administrative subzones to twenty priority hotspots with a three layer evidence trail. The framework is fully parameterised and can be re-run as new amenity data, new census releases or revised weight assumptions become available. We hope that it demonstrates the practical value of combining association rule mining, clustering and supervised learning in a spatially grounded pipeline for urban public health.

Contributions

The work was divided into four balanced tracks so that every member had end to end ownership of a coherent slice of the pipeline. All four members reviewed one another's pull requests, agreed on the configuration defaults and jointly prepared the video script and this report.

Table 9: Contribution summary.

Member	Main areas covered
Angela	Project configuration and shared utility layer, including geospatial helpers, logging and random seed control. Authored the food access and transit feature builders and the spatial choropleth map module used throughout the report.
Esther	Data loading, preprocessing and schema validation for the ten public datasets, together with the container and environment scaffolding. Authored the demographic and land use feature builders and the clustering algorithms with their evaluation metrics and stability analysis.
Clara	Accessibility and binary feature engineering, and the composite feature assembly that produces the modelling table. Authored the Apriori association rule mining module, the documented proxy vulnerability target and the Random Forest risk modelling and interpretation pipeline.
Jiaxuan	Raw geospatial data sourcing and placement, and curation of the twelve item binary matrix configuration. Authored the diagnostic plot library of over twenty figures, the Ward dendrogram subzone labelling enhancement and the end to end pipeline orchestrator.

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